Kevin He

Ph.D. Student in Computer Science
Harvard University
Email: kevinhe@g.harvard.edu
Website: kevinhe.info
LinkedIn: linkedin.com/in/kevin-j-he
Phone: +1 908-300-2878

Education

Sep 2025 – Present **Harvard University**
Ph.D. Student, Computer Science

- **Coursework**: Advanced Computer Architecture, Computing at Scale, Agentic AI for Computer Systems, Machine Learning, Big Data Systems, Foundations of Reinforcement Learning, Advanced Computer Networks (in progress)

Aug 2021 – Dec 2024 **University of California, Berkeley**
Bachelor of Arts, Computer Science GPA: 3.97/4.0

- **Coursework**: Data Structures, Algorithms, Operating Systems & System Programming, Programming Languages & Compilers, Computer Architecture, Digital Design & Integrated Circuits, Advanced Digital Circuits & Systems (Graduate), Hardware for Machine Learning (Graduate), Tapeout, Databases, Networking & Internet Architecture

Fellowships and Honors

2026 **National Science Foundation Graduate Research Fellowship (NSF GRFP)**

2024 **SCALE Scholarship**

Research Interests

Computer architecture and ML systems, with interests in heterogeneous and configurable accelerators, GPU systems, and networking for efficient AI workloads. I'm particularly interested in hardware–software co-design and characterizing compute, memory, and communication costs in system performance and energy efficiency.

Publications

2026 **From Chat to Agents on the Edge: A Cross-Framework Characterization of LLM Inference on Edge GPUs**
Haebin Do, **Kevin He**, David Brooks, and Gu-Yeon Wei
IEEE International Symposium on Workload Characterization (IISWC).

2025 **DEMOTIC: A Differentiable Sampler for Multi-Level Digital Circuits**
Arash Ardakani, Minwoo Kang, **Kevin He**, Qijing Huang, Vighnesh Iyer, Suhong Moon, and John Wawrzynek
Asia and South Pacific Design Automation Conference (ASP-DAC).

2025 **High-Throughput SAT Sampling**
Arash Ardakani, Minwoo Kang, **Kevin He**, Qijing Huang, and John Wawrzynek
Design, Automation and Test in Europe Conference (DATE).

2024 **Late Breaking Results: Differential and Massively Parallel Sampling of SAT Formulas**
Arash Ardakani, Minwoo Kang, **Kevin He**, Vighnesh Iyer, Suhong Moon, and John Wawrzynek
Design Automation Conference (DAC).

Research Experience

Sep 2025 – Present **Harvard Architecture, Circuits and Compilers**
Advised by Prof. David Brooks and Prof. Gu-Yeon Wei

- **PISTIL compiler and accelerator**: Developing compiler support for transformer models on a four-chiplet custom accelerator, including attention, normalization, and activation functions.
- Optimized attention through query-row tiling, on-chip score retention, and recomputation to reduce data movement and scratchpad memory requirements.
- Prototyping accelerator desgins for future chiplet tapeouts
- **GPU Power Optimizations for LLMs**: Characterizing GPU power consumption and inference performance across serving configurations (prefill/decode disaggregation).

Feb 2023 – Jan 2025 **Berkeley SLICE Lab**
Advised by Prof. John Wawrzynek and Prof. Krste Asanović

- **uArchDB**: Built Chisel instrumentation and Python trace analysis tools for debugging CPUs and accelerators in Chipyard. Code / Poster / Slides
- **GNNs for Circuit Power**: Investigated GNN-based power prediction and collected circuit netlists, activity factors, and reference power measurements for training.
- **DiffSampler**: Researched GPU-parallel, gradient-based SAT sampling for design verification; benchmarked sampling uniformity and performance against existing solvers.
- **RISC-V Vector Core Tapeout (Jan–May 2024)**: Co-led a Saturn-with-Rocket tapeout on Intel 16nm; delivered DRC/LVS-clean vector cores at 1 GHz and developed quantized matmul kernels for benchmarking. Documentation / Poster / Slides

Industry Experience

May–Aug 2026 **Architecture Intern**
NVIDIA, Santa Clara, CA

- Developed a hardware-informed fault injection framework in vLLM to study silent data corruption in LLM inference under permanent and intermittent GPU faults
- Integrated GPU kernel profiling and RTL fault distributions to model hardware fault propagation through GEMM and attention operations
- Evaluated LLM fault resilience across dense and MoE models, quantization formats, and prefill/decode phases using reasoning and code generation benchmarks
- Contributed CUTLASS-based attention and grouped GEMM kernels to a GPU diagnostic library

May–Aug 2024 **CPU Performance Intern**
Apple, Cupertino, CA

- Annotated major microarchitecture structures and events in Apple's performance core branch predictors for RTL vs. C++ performance model correlation
- Built new Verilog and performance verification infrastructure for mapping branch predict and train events
- Developed Python tools to parse microarchitecture event logs and quickly extract predictor analytics and identify performance anomalies, significantly reducing initial debugging time to identify issues in waveform dumps

May–Aug 2023 **GPU RTL Design Intern**
Apple, Austin, TX

- Implemented RTL bug fixes in Apple's GPU design library and analyzed performance & area impacts of fixes
- Designed and formally verified a new, reusable, parameterized, low-area SRAM FIFO module for the IP library
- Analyzed performance and area of GPU IP library blocks and developed Python and Tcl scripts to quickly identify configurations and users of library blocks

Teaching

Sep 2026 – Present **Harvard University** — Teaching Assistant
CS2470r: Advanced Topics in Computer Architecture

Jan–May 2024 **UC Berkeley** — Teaching Assistant
EECS151: Introduction to Digital Design

Aug 2022 – Dec 2023 **UC Berkeley** — Instructor
Hands-on PCB Design DeCal

Skills

Python, C/C++, CUDA, PyTorch, Java, SQL, Rust, OCaml, HTML/CSS, RISC-V, x86, ARM, Verilog/SystemVerilog, CHISEL, Git